Clustering Using QUAKES Dataset

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Correlation and Scatterplots

A correlation analysis and scatterplot matrix were used to examine linear relationships among the quantitative variables (*Latitude*, *Longitude*, *Depth*, *Magnitude*, *dNearestStation*, and *RootMeanSquareTime*) within the QUAKES dataset. Figure 1 displays the Pearson correlation coefficients for these variables, and Figure 2 provides the scatterplot matrix. These outputs show that most relationships are weak, meaning the dataset contains limited strong linear associations among the variables.

The geographic coordinates do not show meaningful linear relationships with each other or with most earthquake characteristics. Latitude and Longitude are essentially uncorrelated (r = 0.005), indicating no linear global "east–west vs north–south" pattern in where earthquakes occur. Latitude shows a weak-to-moderate negative relationship with Magnitude (r = -0.281), suggesting that higher-latitude earthquakes in this dataset tend to have slightly lower magnitudes on average. However, the scatterplots indicate substantial spread which could indicate clustering rather than a clean linear trend. Longitude has only weak relationships with Depth (r = -0.046) and Magnitude (r = -0.081), reinforcing that location does not strongly predict earthquake depth or magnitude.

Depth also shows little linear connection to magnitude or measurement variables. Depth has a very weak positive relationship with Magnitude (r = 0.074) and weak negative relationships with dNearestStation (r = -0.152) and RootMeanSquareTime (r = -0.089). Visually, the scatterplots involving Depth show wide dispersion, supporting the conclusion that depth varies broadly and is not strongly tied to the other quantitative variables.

The most notable relationships involve station proximity and measurement quality. The strongest correlation in the table is between dNearestStation and RootMeanSquareTime (r = 0.491), indicating that earthquakes farther from the nearest station tend to have higher timing errors. Longitude is also moderately correlated with dNearestStation (r = 0.334) and RootMeanSquareTime (r = 0.409), which suggests that station coverage and measurement quality may vary by geographic region. Magnitude shows weak-to-moderate positive relationships with dNearestStation (r = 0.284) and RootMeanSquareTime (r = 0.320). This indicates that larger events are somewhat associated with greater station distance and higher timing error, though these relationships remain too weak to support strong prediction.

Figure 1

Correlations among Quantitative Variables

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc corr data=SASHELP.QUAKES pearson nosimple noprob plots=none;
    var Latitude Longitude Depth Magnitude dNearestStation RootMeanSquareTime;
run;
```

The current time is 12:55:56 and the date is 15FEB2026

6 Variables:	Latitude Longitude Depth Magnitude dNearestStation RootMeanSquareTime

Pearson Correlation Coefficients
Number of Observations

	Latitude	Longitude	Depth	Magnitude	dNearestStation	RootMeanSquareTime
Latitude	1.00000 15578	0.00512 15578	-0.00594 15578	-0.28095 15578	0.03472 3664	-0.16858 6289
Longitude	0.00512 15578	1.00000 15578	-0.04654 15578	-0.08107 15578	0.33418 3664	0.40866 6289
Depth	-0.00594 15578	-0.04654 15578	1.00000 15578	0.07406 15578	-0.15179 3664	-0.06951 6289
Magnitude	-0.28095 15578	-0.08107 15578	0.07406 15578	1.00000 15578	0.28411 3664	0.32021 6289
dNearestStation	0.03472 3664	0.33418 3664	-0.15179 3664	0.28411 3664	1.00000 3664	0.49909 3660
RootMeanSquareTime	-0.16858 6289	0.40866 6289	-0.06951 6289	0.32021 6289	0.49909 3660	1.00000 6289

Note. This figure shows the correlations among the quantitative variables (*Latitude*, *Longitude*, *Depth*, *Magnitude*, *dNearestStation*, and *RootMeanSquareTime*) within the QUAKES dataset. It also provides the source code used to generate the output.

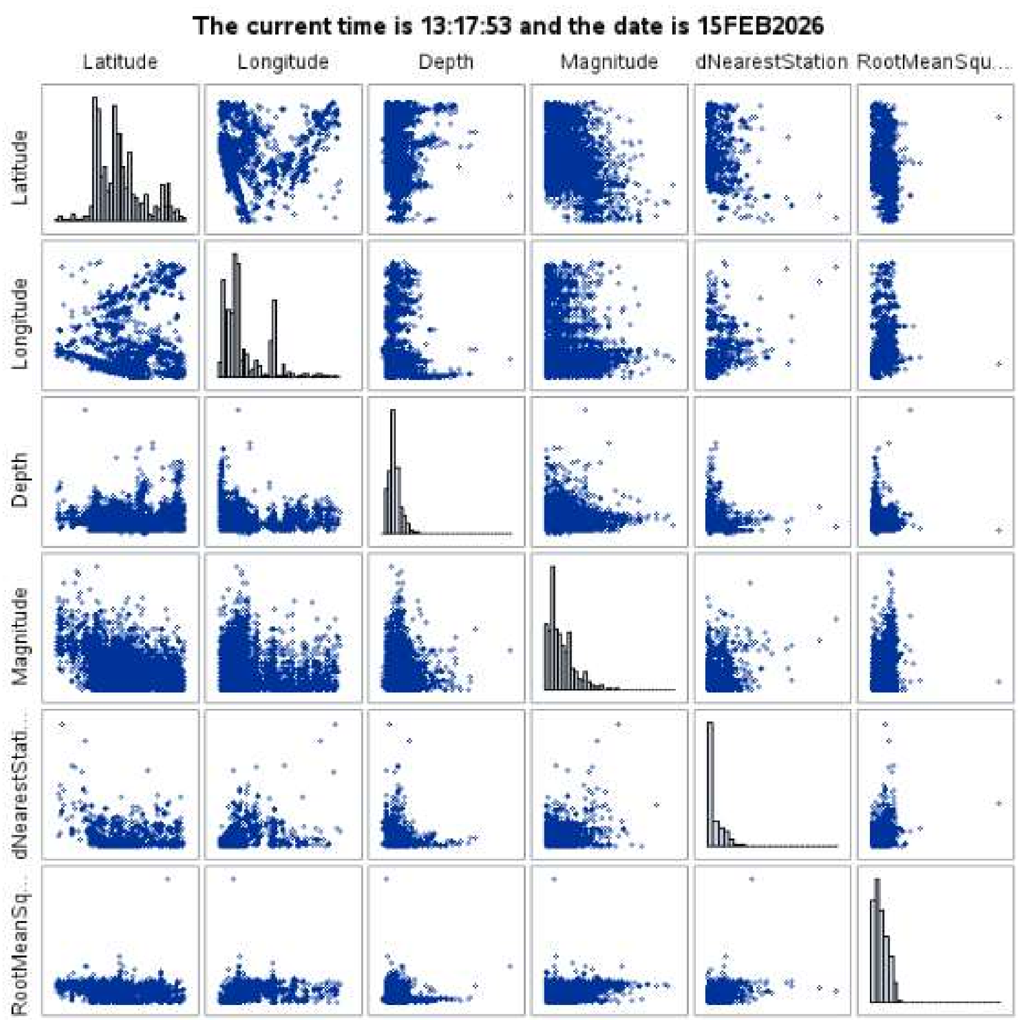
Figure 2

Scatterplot Matrix for the Quantitative Variables

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc sgscatter data=sashelp.quakes;
   matrix Latitude Longitude Depth Magnitude dNearestStation RootMeanSquareTime /
          diagonal=(histogram);
run;
```



Note. This figure shows the scatterplots for the quantitative variables (*Latitude*, *Longitude*, *Depth*, *Magnitude*, *dNearestStation*, and *RootMeanSquareTime*) within the QUAKES dataset. The diagonals within the matrix are histograms of the variables. It also provides the source code used to generate the output.

Summary Statistics

Figure 3 is the summary statistics for the quantitative variables. Latitude, Longitude, Depth, and Magnitude each contain 15,578 observations, while DNearestStation (3,664) and RootMeanSquareTime (6,289) have substantially fewer. This indicates a large amount of missing data in those two variables. This is an important characteristic as it may affect later analysis if missing values reduce the usable sample size.

The geographic variables show broad coverage, with Latitude ranging from 25.01 to 48.99 (mean = 36.86) and Longitude ranging from -123.99 to -64.54 (mean = -111.83). The standard deviation for these coordinates are relatively large which reinforces that earthquake events are spread across many geographic regions rather than concentrated in a narrow band of coordinates.

Depth and Magnitude provide insight into earthquake characteristics and potential data quality issues. Depth ranges from -3.30 to 104.30 with a mean of 7.06. The negative minimum depth is not realistic for earthquakes and suggests measurement error. Magnitude ranges from 2.50 to 7.20 with a mean of 3.07. This indicates that most earthquakes in the dataset are relatively small to moderate in magnitude, with fewer high-magnitude events. This is consistent with typical earthquake patterns where smaller events occur more frequently than large earthquakes.

Finally, the monitoring-related variables suggest differences in measurement conditions across observations. dNearestStation ranges from 0 to 5.90 with a mean of 0.32, meaning that many earthquakes were recorded relatively close to a monitoring station, RootMeanSquareTime ranges from 0 to 8.46 with a mean of 0.53, showing that most observations have low timing error, but a small number of earthquakes have substantially higher error. Overall, key characteristics of the dataset include broad geographic coverage, a magnitude distribution dominated by smaller events, a potential depth data quality concern, and significant missingness in station-distance and timing variables, which may affect the clustering analysis.

Figure 3

Summary Statistics for Quantitative Vairables

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc means data=SASHELP.QUAKES chartype mean std min max n vardef=df;
    var Latitude Longitude Depth Magnitude dNearestStation RootMeanSquareTime;
run;
```

The current time is 14:19:08 and the date is 15FEB2026

Variable	Mean	Std Dev	Minimum	Maximum	N
Latitude	36.8590070	4.5585671	25.0130000	48.9900000	15578
Longitude	-111.8298258	10.9945729	-123.9980000	-84.5386000	15578
Depth	7.0634769	5.8752128	-3.3000000	104.3000000	15578
Magnitude	3.0694184	0.5084512	2.5000000	7.2000000	15578
dNearestStation	0.3184727	0.4146903	0	5.8960000	3664
RootMeanSquareTime	0.5301682	0.4360713	0	8.4600000	6289

Note. This figure shows the summary statistics for the quantitative variables (*Latitude*, *Longitude*, *Depth*, *Magnitude*, *dNearestStation*, and *RootMeanSquareTime*) within the QUAKES dataset. It also provides the source code used to generate the output.

Graphical Summary

The diagonal panels of the scatterplot matrix (Figure 2) provide a graphical summary (histograms) for each quantitative variable. The histograms for Latitude and Longitude show that the earthquake locations are not evenly distributed and appear concentrated in certain coordinate ranges which suggest geographic clustering. Depth shows a strongly right-skewed distribution, with most earthquakes occurring at relatively shallow depths and fewer events occurring at greater depths. Magnitude is also right-skewed, with most earthquakes falling in the lower magnitude range and relatively few high-magnitude events. The dNearestStation histogram shows a heavy concentration near smaller values with a long right tail, indicating that most earthquakes were recorded close to a monitoring station. RootMeanSquareTime is similarly right-skewed, with most earthquakes having low timing errors.

Cluster Analysis

FASTCLUS was selected because it clusters earthquake observations based on the quantitative variables, which directly supports the goal of identifying geographic and severity-based groupings relevant to emergency service planning. Because the QUAKES dataset contains substantial missing data for dNearestStation and RootMeanSquareTime, the FASTCLUS analysis was divided into two separate clustering runs. Latitude, Longitude, Depth, and Magnitude contain complete data for all 15,578 earthquakes. This allows clustering to be performed using the full dataset and ensuring that the geographic and earthquake patterns were not distorted by missingness. In contrast, including dNearestStation and RootMeanSquareTime requires using only complete-case observations. It reduces the usable sample size considerably (to 3,660 observations) and changes the interpretation of the clusters because the results then represent only the subset of earthquakes with station-distance and RootMmeanSquareTime information available. For this reason, a core-variable clustering model was run first to capture the broad clustering structure of the full dataset, followed by an all-variable clustering model to evaluate whether station proximity and measurement quality meaningfully change the cluster patterns.

Figure 4 presents the results for the core-variable model using standardized Latitude, Longitude, Depth, and Magnitude. Multiple cluster solutions were evaluated (k = 4 through 6), and the k = 5 solution was selected as it produced the strongest overall clustering performance. The cluster summary shows that the five clusters are uneven in size, with Cluster 2 representing the largest group of earthquakes (n = 6,626) and Cluster 1 representing the smallest group (n = 491). The overall R-square value (0.553) and pseudo-F statistic (5281.15) indicate meaningful separation between clusters. The variable statistics show that Longitude has the highest R-square (0.7488) in Cluster 4, suggesting that geographic location contributes strongly to how the clusters are formed. The cluster means further show that clusters differ most noticeably by location and depth, while Magnitude contributes to separation for at least one cluster. The core clustering solution reflects both geographic concentration and differences in earthquake characteristics.

Figure 4

FASTCLUS (k=5): Latitude, Longitude, Depth, Magnitude

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc stdize data=sashelp.quakes out=quakes_core_z method=std;
  var Latitude Longitude Depth Magnitude;
run;

proc fastclus data=quakes_core_z maxclusters=5 maxiter=100 out=core_clusters;
  var Latitude Longitude Depth Magnitude;
run;
```

The current time is 15:54:33 and the date is 15FEB2026

Cluster Summary

Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	491	1.2051	13.4855		2	3.6976
2	6626	0.5378	2.4752		5	1.9137
3	2319	0.7523	6.4029		2	2.1273
4	3378	0.6782	5.3221		2	2.2160
5	2764	0.6360	3.0478		2	1.9137

Statistics for Variables

Variable	Total STD	Within STD	R-Square	RSQ/(1-RSQ)
Latitude	1.00000	0.64627	0.582449	1.394915
Longitude	1.00000	0.50176	0.748307	2.973087
Depth	1.00000	0.76261	0.418573	0.719908
Magnitude	1.00000	0.66849	0.553230	1.238288
OVER-ALL	1.00000	0.65151	0.575640	1.356488

Pseudo F Statistic =	5281.15

Approximate Expected Over-All R-Squared =	0.55300

Cubic Clustering Criterion =	18.635

WARNING: The two values above are invalid for correlated variables.

Cluster Means

Cluster	Latitude	Longitude	Depth	Magnitude
1	1.058321755	-0.702853001	3.397331804	-0.202221225
2	-0.380351313	-0.545198494	-0.004480656	-0.273736111
3	-0.851702048	-0.245247151	0.153763083	1.772819520
4	-0.016621740	1.628691065	-0.183393436	-0.416812380
5	1.458689257	-0.352895887	-0.497639165	-0.285857508

Cluster Standard Deviations

Cluster	Latitude	Longitude	Depth	Magnitude
1	1.175582140	0.601481918	1.836356350	0.832646745
2	0.551029524	0.279087863	0.711783360	0.518252208
3	0.762017526	0.493357568	0.729750686	0.952359962
4	0.674558981	0.748855832	0.622282792	0.660980348
5	0.581042805	0.527683991	0.738263446	0.675952399

Note. Output from FASTCLUS (k-mean clustering) using standardized earthquake variables *Latitude, Longitude, Depth, and Magnitude* from the QUAKES dataset. It also provides the source code used to generate the output.

Although k = 5 produced the highest pseudo F statistic, the k = 6 solution was selected because it produced the highest overall R-square and resulted in a more stable and interpretable cluster structure without missing cluster summary values. The k = 6 model also reduced dominance by a single large cluster and better separated observations with extreme RootMeanSquareTime values.

Figure 5 presents the FASTCLUS results for the all variable clustering model using standardized Latitude, Longitude, Depth, Magnitude, dNearestStation, and RootMeanSquareTime. To maintain consistency with the core-variable clustering model, a k = 5 solution was selected for this analysis as well. Because dNearestStation and RootMeanSquareTime contain substantial missing data in the dataset, this clustering solution represents only the complete-case subset of observations rather than the full dataset. The cluster summary shows that the five clusters are uneven in size, with Cluster 2 representing the largest group of earthquakes (n = 7,376) and Cluster 5 representing the smallest group (n = 781). The overall R-square value (0.369) and pseudo F statistic (4088.06) indicates moderate separation between clusters. This implies that meaningful structure exists in the data even after including the additional monitoring-related variables. The variable statistics indicate that Longitude contributes most strongly to cluster separation (R-square = 0.694), followed by RootMeanSquareTime (R-square = 0.611) and Magnitude (R-square = 0.497. Latitude contributes the least (R-square = 0.408). The cluster means further show that the five clusters differ primarily by geographic location and measurement-related characteristics, with one cluster (Cluster 4) standing out due to an extremely high RootMeanSquareTime (18.185), suggesting that it captures earthquakes with unusually large timing residuals and potentially lower-quality measurement conditions

Figure 5

FASTCLUS (k=5): All Variables

```
data _null_;
call symput ('timenow',put (time(),time.));
call symput ('datenow',put (date(),date9.));
run;

title "The current time is &timenow and the date is &datenow";

proc stdize data=sashelp.quakes out=quakes_all method=std;
  var Latitude Longitude Depth Magnitude dNearestStation RootMeanSquareTime;
run;

proc fastclus data=quakes_all maxclusters=4 maxiter=100 out=all_clusters;
  var Latitude Longitude Depth Magnitude dNearestStation RootMeanSquareTime;
run;
```

The current time is 16:41:19 and the date is 15FEB2026

Cluster Summary

Cluster	Frequency	RMS Std Deviation	Maximum Distance from Seed to Observation	Radius Exceeded	Nearest Cluster	Distance Between Cluster Centroids
1	3968	0.8019	6.5279		2	2.5678
2	7376	0.5461	4.2959		1	2.5678
3	2341	1.0758	12.4045		1	3.4218
4	1112	.	3.6961		3	17.5882
5	781	0.9934	16.3966		2	3.0263

Statistics for Variables

Variable	Total STD	Within STD	R-Square	RSQ/(1-RSQ)
Latitude	1.00000	0.76929	0.408342	0.690166
Longitude	1.00000	0.55347	0.693753	2.265343
Depth	1.00000	0.75623	0.428260	0.749048
Magnitude	1.00000	0.70927	0.497068	0.988341
dNearestStation	1.00000	0.75294	0.433708	0.765872
RootMeanSquareTime	1.00000	0.62396	0.610924	1.570194
OVER-ALL	1.00000	0.69854	0.512204	1.050037

Pseudo F Statistic =	4088.06

Approximate Expected Over-All R-Squared =	0.36883

Cubic Clustering Criterion =	150.056

WARNING: The two values above are invalid for correlated variables.

Cluster Means

Cluster	Latitude	Longitude	Depth	Magnitude	dNearestStation	RootMeanSquareTime
1	0.15843549	1.41017156	-0.26579561	-0.37776417	0.65185275	0.45967221
2	-0.16609250	-0.55270312	-0.12251018	-0.24630329	-0.34758942	-0.80476723
3	-0.87046837	-0.20991132	0.14115890	1.67103894	2.33049641	1.37843875
4	1.87048393	-0.51626010	-0.47920910	-0.35531316	6.17709467	18.18471228
5	0.70962177	-0.58045955	2.76663205	-0.25748091	-0.23152243	-0.63131932

Cluster Standard Deviations

Cluster	Latitude	Longitude	Depth	Magnitude	dNearestStation	RootMeanSquareTime
1	0.804899833	0.859240404	0.619954988	0.675623400	1.028307050	0.757698533
2	0.724944261	0.303130071	0.670319563	0.607753233	0.464647628	0.370927154
3	0.777166269	0.512866775	0.693014989	1.012954916	1.995419117	0.767738741
4	0.397274652	0.455022490	0.803596086	0.618826992	2.745097296	.
5	1.230809837	0.638903148	1.689416582	0.778584406	0.491722565	0.543793585

Note. Output from FASTCLUS (k-mean clustering) using standardized earthquake for all variables from the QUAKES dataset. It also provides the source code used to generate the output.

Conclusion

These results suggest that the QUAKES dataset contains meaningful geographic clustering that supports the use of Latitude and Longitude for identifying regions where earthquake activity is concentrated. While most quantitative variables do not show strong linear relationships, the FASTCLUS results demonstrate that earthquakes can be grouped into distinct clusters based on location and event characteristics. Including monitoring-related variables (dNearestStation and RootMeanSquareTime) further differentiates clusters by measurement conditions. Taken together, these findings indicate the dataset is suitable for use in the next phase of the organization's analytics project and provide a reasonable basis for evaluating where emergency services may be most needed.